Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University, Solapur

Choice Based Credit System (CBCS)
Structure and Syllabus
Multidisciplinary Minor Program in
Electronics and Computer Engineering
offered by
Department of Electronics Engineering
for Batch Admitted in
2025-26

Dean (Academics)

Walchand Institute of Technology Electronics Dept. Solapur.

HEAD
Electronics Department
Walchand Institute of Technology
SOLAPUR-413006.

Multidisciplinary Minor Program in Electronics and Computer Engineering	
Legends Used	
L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B.Tech.	Bachelor of Technology

Walchand Institute of Technology Electronics Dept. Solapur

Multidisciplinary Minor Program in Electronics and Computer Engineering

Course Code Format

2	4	E	C	U/P	2	C	C	1	T/L
Year of syllabus revision		Program Code		U-Under Graduate, P-Post Graduate	Semester No./ Year1/2/3/…8	Course Type		Course Serial No. 1-9	T-Theory, L-Lab session, A-Tutorial, P-Programming / Design

Program Code	
ET	Electronics and Telecommunication Engineering
Course Type	
BS	Basic Science
ES	Engineering Science
HU	Humanities & Social Science
MC	Mandatory Course
CC	Programme Core Compulsory Course
SN*	Self-Learning (**N*** *indicates the serial number of electives offered in the respective category)*
EN*	Programme Core Elective Course (**N*** *indicates the serial number of electives offered in the respective category)*
SK	Skill-Based Course
SM	Seminar
MP	Mini project
PR	Project
IN	Internship
ON*	Open Elective (**N*** *indicates the serial number of electives offered in the respective category)*
MD	Multidisciplinary Minor

EM	Entrepreneurship/Economics/Management Courses
FP	Community Engagement Project / Field Project
AE	Ability Enhancement Course
VE	Value Education Course
IK	Indian Knowledge System
VS	Vocational & Skill Enhancement Course
RM	Research Methodology
HN	Honors' Degree Course
HR	Honors Research

Sample Course Code	
25ECU3MD6T	Analog Electronics

Walchand Institute of Technology
Electronics Dept.
Solapur.

Multidisciplinary Minor Program in Electronics and Computer Engineering

Semester	Course Code	Name of Course	Engagement Hours			Credits					Total
			L	T	P		Theory	OE/ POE	ISE	ICA	
III	25ECU3MD6T	Analog Electronics	2	-	-	2	60	-	40	-	100
IV	25ECU4MD6T	Digital Electronics	1	-	-	1		-	50	-	50
V	25ECU5MD6T	Microcontrollers and Peripherals	3	-	-	3	60	-	40	-	100
VI	25ECU6MD6T	Advanced Controllers and Interfacing	2	-	-	2	60	-	40	-	100
VII	25ECU7MD6T	Electronic System Design	3	-	-	3	60	-	40	-	100
VII	25ECU7MD6A	Electronic System Design(Tutorial)	-	1	-	1	-	-	-	25	25
		Sub-Total	**11**	**1**	**0**	**12**	**240**	**-**	**210**	**25**	**475**
Laboratory Courses											
IV	25ECU4MD6L	Digital Electronics	-	-	2	1	-	-	-	25	25
VI	25ECU6MD6L	Advanced Controllers and Interfacing	-	-	2	1	-	-	-	25	25
		Sub-Total	**0**	**0**	**4**	**2**	**0**	**-**	**0**	**50**	**50**
		Grand-Total	**11**	**1**	**4**	**14**	**240**	**-**	**210**	**75**	**525**

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Walchand Institute of Technology Solapur
Electronics Dept.

	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **Second Year B.Tech. (Multidisciplinary Minor in ECM Engineering, Semester-III**

25ECU3MD6T : Multidisciplinary Minor I –Analog Electronics

Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	60 Marks
Credits	2	**ISE**	40 Marks

Introduction:

This program basic course provides essential fundamental knowledge to the student to understand working of electronic devices, their analysis and designing of electronic applications using these electronic devices. It also covers design and analysis applications IC 555 timer and IC 741.This is one of the foundation courses which is vital for students to comprehend working of complex electronic circuits and systems.

Course Prerequisite:

The student needs to complete an introductory course in basic electronics which includes basic knowledge of diode, rectifiers, BJT and different BJT configurations. The student is expected to apply basic electrical theorems such as Kirchhoff's current law and Thevenin's theorem in electrical circuits to analyze the circuit performance.

Course Objectives:

1. To introduce concept of regulated and unregulated dc power supplies
2. To understand biasing techniques and applications of bipolar junction transistors.
3. To make aware Op-Amp characteristics, parameters, and various linear applications in analog signal processing
4. To understand the functioning of timer IC 555 and its applications.

Course Outcomes:

After completing this course, the student will be able to -

1. Differentiate between regulated and unregulated DC power supplies
2. Describe BJT biasing methods and their uses in circuits
3. Describe the characteristics and parameters of operational amplifiers and utilize them in linear analog signal processing circuits
4. Explain the working of timer IC 555 and make its use in various timing and pulse generation applications

<table>
<tr><td>Unit – I</td><td>Regulated power supply</td><td>05 Hours</td></tr>
<tr><td colspan="3">Bridge rectifier, analysis of unregulated dc power supply using bridge rectifier and shunt capacitor filter. Regulated dc power supply, three pin regulators, construction of 5V and 12V regulated dc power supply using capacitor filter and three pin regulator IC 7805, 7812.</td></tr>
<tr><td>Unit – II</td><td>Bipolar Junction Transistors (BJT) Application</td><td>05 Hours</td></tr>
<tr><td colspan="3">Comparison of BJT configurations, BJT application as an amplifier - Voltage divider biasing, DC load line, operating point, thermal runaway, single stage CE amplifier , voltage gain and frequency response, BJT application as switch.</td></tr>
<tr><td>Unit – III</td><td>Op-Amp and its applications</td><td>05 Hours</td></tr>
<tr><td colspan="3">Op-Amp block diagram, and characteristics of Op-Amp, equivalent circuit of Op-Amp, practical Op-Amp, comparison of ideal and practical Op-Amp, Op-Amp parameters- input offset voltage, output offset voltage, input bias current, common mode rejection ratio, slew rate, power supply rejection ration (PSRR).
Op-Amp applications- Inverting, non-inverting amplifier, adder, subtractor, summing- scaling - averaging amplifier, integrator, differentiator, comparator.</td></tr>
<tr><td>Unit – IV</td><td>IC555 Timer</td><td>05 Hours</td></tr>
<tr><td colspan="3">IC 555- pin diagram, internal block diagram, Astable and Monostable multivibrator design using IC 555, design of timer application.</td></tr>
<tr><td colspan="3">Text Books</td></tr>
<tr><td colspan="3">1. Op-Amps and Linear Integrated Circuits, Ramakant A. Gayakwad, PHI Learning Pvt. Ltd., Fourth edition
2. Electronic Devices and Circuits, Allen Mottershed, PHI Publication
3. Electronic Devices and Circuits, David A.Bell, Oxford University, Press India, Fifth edition.</td></tr>
<tr><td colspan="3">Reference Books</td></tr>
<tr><td colspan="3">1. An introduction to Operational Amplifiers, Lucas M. Faulkenberry, John Wiley& Sons, Second edition.
2. Operational Amplifiers, G. B. Clayton,English Language Book Society, Second edition.
3. Electronic Devices and Circuits, Robert Boylestad, Prentice Hall International.
4. Electronic Circuit Design, S.N.Talbar, T.R.Sontakke, Sadhu Sudha Publications.</td></tr>
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WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Second Year B.Tech. (Multidisciplinary Minor in ECM Engineering), Semester-IV

25ECU4MD6T : Multidisciplinary Minor II – Digital Electronics

Teaching Scheme		Examination Scheme	
Lectures	1 Hours/week	**ISE**	50 Marks
Practical	2 Hours/week	**ICA**	25 Marks
Credits	2		

Introduction:

Digitization has spread to a wide range of applications, including information (computers), automation, communications, control systems and signal processing. This course on digital electronics provides a thorough understanding of digital system design. The course intends to cover combinational and sequential circuit analysis and design.

Course Prerequisite:

Student shall have knowledge of binary number system and logic gates. Student shall also have knowledge about basic electronics devices – diodes, BJT and FET.

Course Objectives:

1. To introduce to student concepts of digital logic circuits like number systems, Boolean algebra, operation of various gates etc. along with their applications.
2. To make student design combinational circuit.
3. To make student design sequential circuit.

Course Outcomes:

After completing this course, student will be able to –

1. Perform operations on BCD, gray codes and error- correcting codes (e.g. hamming code) for digital communication and error detection application.
2. Differentiate between logic families based on various performance characteristics
3. Simplify and implement Boolean functions using Boolean algebra, DeMorgan's laws, karnaugh maps and canonical forms (SOP, POS), while applying optimal gate-level design techniques using NAND & NOR gates.
4. Design and implement combinational circuits(e.g. adders, multiplexers, decoders) and sequential circuits(e.g. counters, registers, flip-flops) by utilizing appropriate design techniques

Unit – I	Number System and Codes	04 Hours

Review of number system, BCD codes, gray code, representation of signed numbers, error detecting and correcting codes – parity codes and hamming code, introduction to logic families, parameter definitions – noise margin, power dissipation, fan-in, fan-out, propagation delay, typical values for TTL, CMOS&ECL.

Unit – II	Boolean Algebra and Logic Simplification	04 Hours
Theorems of Boolean algebra, DeMorgan's law, standard representation of logic functions – SOP, POS and canonical forms, simplification of logic functions – karnaugh maps, NAND and NOR implementations.		
Unit – III	**Combinational Circuit Design**	**03 Hours**
Design and analysis procedure, design of adder, subtractor and binary parallel adder, code conversion, design of multiplexers, demultiplexers, encoders, decoders and their applications, design of magnitude comparators and parity circuits.		
Unit – IV	**Sequential Circuit Design**	**04 Hours**
Latches - SR latch; Flip-flops: SR, JK, D, T and Master-slave, triggering of flip-flops, Flip-flop characteristic equations and excitation tables. Flip-flop applications –counters, registers		

Internal Continuous Assessment (ICA):

- ICA shall consist of minimum 8 experiments based upon-
  1. Verification of truth table of basic and universal gates.
  2. Implementation of universal gates using basic gates.
  3. Code conversion using logic gates: binary to gray, gray to binary, binary to excess-3.
  4. Implementation of any one combinational circuit using multiplexer.
  5. Implementation of any one combinational circuit using de-multiplexer.
  6. Design a n-bit comparator using logic gates.
  7. Perform 1's and 2's complement adder/subtraction using 4 bit parallel adder.
  8. Convert J-K flip-flop to T (Toggle) flip-flop and D (Data) flip-flop.
  9. Design and implement mod-n asynchronous counter.
  10. Design and implement mod-n synchronous counter.
  11. Design and implement a 4 bit bi-directional shift register.

Text Books

1. Digital Fundamentals, Thomas L.Floyd, Global Edition-Pearson Education Limited
2. Fundamental of Digital Circuits-Anand Kumar-Prentice Hall of India Pvt.Ltd.
3. Digital Design, M.Morris Mano, PHI,Third edition
4. Modern Digital Electronics, R.P.Jain, TMH,Third edition

Reference Books

1. Digital Design: Principles and Practices, Wakerly JohnF., Pearson Education, Forth edition
2. Digital Principles &Applications, Leach, Malvino, Sixth edition